Deep Learning Segmentation of Invasive Melanoma

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Paper ID 224

Abstract

Melanoma is the deadliest skin cancer with the fastest rising incidence rate in the United States. The most important predictor of melanoma patient survival is the volume of invasive tumor at the initial biopsy. The appearance of in-situ melanoma and invasive melanoma, which invades the underlying soft tissue and drives mortality, is visually the same. The current standard of care for outcome prediction is to manually measure a single-dimension tumor thickness as a proxy for volume. The predictive accuracy of this single measurement depends on a host of subjectively-defined measurements that can lead to incorrect risk assessment and management planning. Therefore, in order to mitigate subjective measurements we propose a novel two-stage method. The first stage computes segmentation maps for melanoma and epidermis. The second stage combines melanoma and epidermis predictions from the first stage to yield invasive melanoma predictions. Our method utilizes multiple resolutions and downscaling to increase information passed to the model and to improve model accuracy. Using an HRNet+OCR model for both epidermis and melanoma segmentation in our proposed system results in a marked improvement of F1 score (mIoU) to 0.44 (0.64) as compared to the current state-of-the-art of 0.14 (0.53).

1. Introduction

Melanoma is a lethal cancer with the fastest rising incidence of any cancer in the United States [1]. The national SEER cancer incidence data estimates 100,350 new cases in 2020 [2]. As a skin cancer, it can invade through the first layer of the skin, called the epidermis, into the dermis layer below. Melanoma within the epidermis, called in-situ melanoma, is not known to cause death [3]. However, once melanoma is in the dermis, called invasive melanoma, the tumor cells could travel to distant body sites, in a fatal process called metastasis. The primary invasive tumor burden is a crucial factor for patient survival prediction (staging) and clinical management [4]. Therefore, accurate assessment of the invasive tumor burden is paramount. Over-staging subjects patients to unnecessary risks from interventions such as sentinel lymph node biopsy, whole-body PET/CT, and treatments, and also adds an undue financial burden on the health care system. Under-staging would potentially delay needed care for the patient.

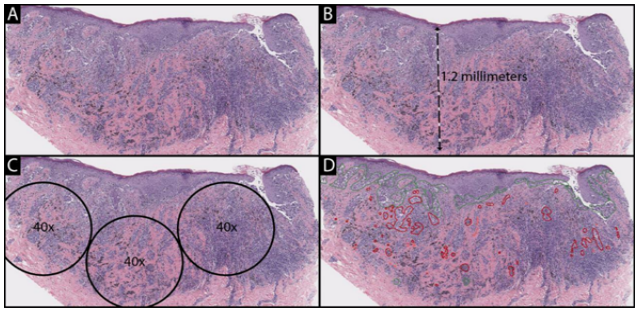


Figure 1: **(a)** the cross-section of a melanoma; **(b)** the Breslow thickness is measured from the granular layer to the deepest aspect of the cancer; **(c)** manual area estimation method [5] that counts the number of tumor-occupying microscopic fields, shown in the black circles, combined with tumor density estimation, and **(d)** areas of invasive melanoma in red circles

In current practice, the invasive tumor size is assessed using Breslow thickness, a measurement developed in 1970 by Dr. Alexander Breslow, from a cross-sectional image [6]. However, this one-dimensional measurement has several limitations. Breslow et. al. reported that the incidence of metastasis is a function of the cross-sectional area, rather than a function of the thickness alone [6]. Saldanha et al. reported cross-sectional area to be an independent prognostic factor for survival [5].

To overcome the limitations of Breslow thickness, Saldanha et al. [5] proposed a manual method that estimates the invasive tumor cross-sectional area, shown in Figure 1, which better predicts mortality than Breslow thickness. However, this manual method is time-intensive with high inter-observer variability, thereby limiting its clinical utility and adoption [4]. Deep learning based semantic segmentation can potentially offer an accurate and efficient al-

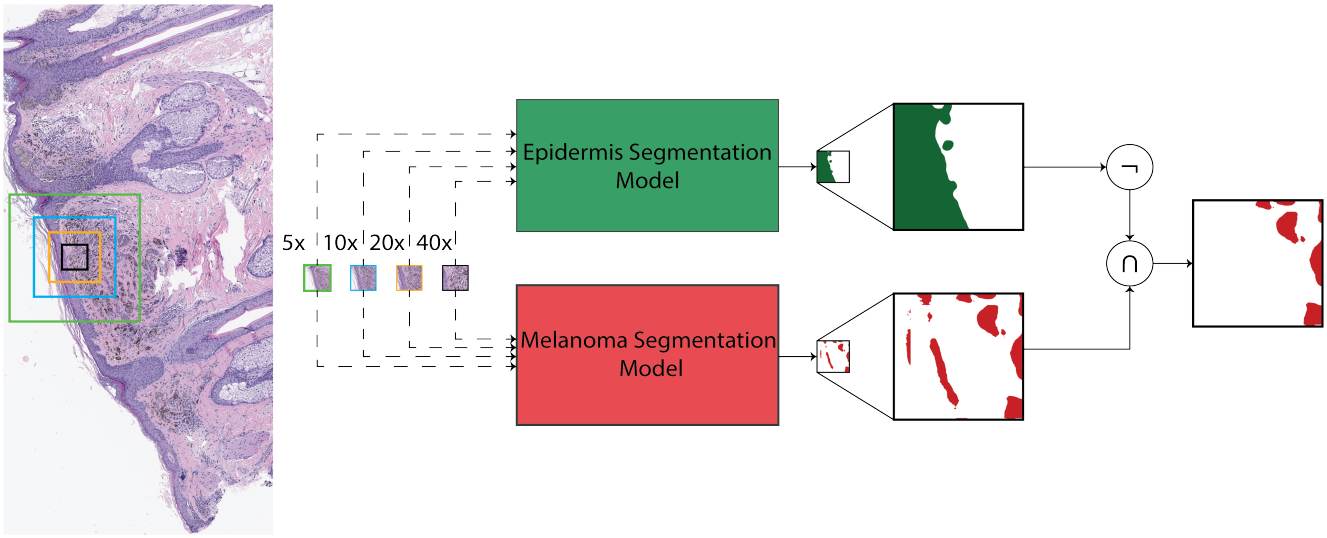


Figure 2: Block diagram of our proposed two-stage method. In the above image patches, the green, blue, orange, and black bounding boxes correspond to $5\times$, $10\times$, $20\times$, and $40\times$ magnifications that are all downscaled to match the $40\times$ patch size. Depending on model used, some subset of downscaled patches are passed into the models as indicated by the dashed lines. After the epidermis segmentation predictions are computed the outputs are negated and intersected with the outputs of the melanoma segmentation model to create invasive melanoma predictions.

ternative for evaluating the invasive tumor cross-sectional area. Automated tumor area calculation has been shown to precisely identify tumor occupying regions for breast and prostate cancers in the works of Su et al. [7] and Burlutskiy et al. [8] which supports our application to melanoma. This approach is also highly reproducible, which is necessary for practical adoption. Thus, deep learning based methods would be invaluable for staging and management planning, and can significantly impact the standard of care.

In this paper, we introduce a novel deep learning method to segment invasive melanoma. Unlike previous works [9, 10] which primarily aim to segment large tumors, we aim to segment tumors of all sizes; notably small tumors that contain single cells and small clusters of melanoma. Two resulting consequences of this make our task more difficult than prior work. First, our method must perform well while having increased clinical applicability; second, there is an exaggerated class imbalance. Our data is annotated by board-certified dermatopathologists to the cellular and pixel level, which assures high quality data for training and testing. Since in-situ melanoma and invasive melanoma are visually indistinguishable, we propose a two-stage system to distinguish between the two, as shown in Figure 2. In the first stage we perform two concurrent steps: semantic segmentation of the epidermis and semantic segmentation of melanoma. In the second stage, we use these two segmentation maps to classify invasive melanoma as the intersection of negative epidermis and positive melanoma predictions. In doing so, our method aims to remove false positive predictions of invasive melanoma in the epidermis. We also leverage methods such as detail-preserving image downscaling for preprocessing, that incorporate necessary information from larger areas of an image while preserving high-resolution information.

2. Related Work

In 1999, LeCun et al. demonstrated that a convolutional neural network (CNN) could learn to extract the right features from pictorial objects with high shape variability [11]. A CNN was able to achieve the tasks with minimal human intervention in image preprocessing. This has led to dramatic advancements in many areas of healthcare, particularly in pathology image analysis. Su et al. segmented breast cancer with a sensitivity of 0.91, specificity of 0.82, and F1 score of 0.85 [7]. Burlutskiy et al. achieved a F1 score of 0.80 on segmenting potentially cancerous areas in prostate biopsies [8]. Bulten et al. also segmented for prostate cancer, with a reported F1 score of 0.876 [12]. These prior studies demonstrate that CNNs can be an effective method for the task of cancer segmentation for microscopic images.

Several recent studies in 2019 and 2020 have reported the use of computer vision for melanoma segmentation. These studies were limited by their accuracy, clinical utility, and generalizability. Van Zon et al. segmented melanoma for tumor segmentation of benign nevus versus melanoma [9]. With 563 cases, their U-Net CNN achieved an accuracy of 0.92 and F1 score of 0.84. However, their segmentation

method did not distinguish in-situ melanoma from invasive melanoma and therefore cannot be used in clinical practice. Phillips et al. segmented the epidermis and melanoma in the dermis using an original multi-stride fully convolutional network, but their training and testing data was limited to large nodular melanomas, which are rare in clinical practice [10]. Most melanoma tumors are much smaller, some with an invasive component of scattered single cells and small clusters of cells. Furthermore, their cases were hand selected from The Cancer Genome Atlas to avoid variations in slide quality, which limits the generalizability of their trained algorithm. Their algorithm achieved a mean pixel accuracy of 0.624 [10]. The algorithm trained by Kulkarni et al. used segmentation to predict metastasis. Their method utilized a combination of traditional machine learning, CNNs, and recurrent neural-networks in an unsupervised manner [13]. Their tumor segmentation performance was not reported, but it appeared to misidentify adnexa, vessels, and collagen bundles as melanoma in their example figure. Their method also did not distinguish melanoma in the epidermis from melanoma in the dermis.

Outside of melanoma segmentation, we draw inspiration from existing literature for ideas addressing problems in whole slide image (WSI) processing and tumor segmentation. Due to the large size of WSIs, they must be divided into smaller images, called patches, in order to be fed into a neural network, and necessary information is often lost in this process. To address this issue, Gu et al. developed the Multi-Resolution Network in 2018 [14]. This network consisted of a U-Net model and an additional encoder pathway which mimicked the structure of the first half of their U-Net architecture. A patch at the resolution being used for segmentation was fed into the U-Net model, and a lower resolution patch was fed into the additional encoder pathway. The outputs at each step of the additional encoder pathway were cropped, resized, and then concatenated onto the corresponding depth of the U-Net decoder pathway. From ROC curves generated on their test set, it was deduced that the Multi-Resolution Network led to performance gains over the single-resolution U-Net.

In addition, Reinforced Auto-Zoom Net by Dong et al. in 2018 utilized reinforcement learning to develop a policy that determines the optimal segmentation resolution for each patch of a given WSI [15]. The policy has two output actions: increase resolution or "zoom-in", or initiate segmentation at the current resolution. The use of this policy led to significant mIoU gains when compared to the single-resolution network, DeepLab. Specifically, the DeepLab network attained an mIoU of 0.18 while the Reinforced Auto-Zoom Net attained an mIoU of 0.49 on a test set of breast cancer WSIs. Building on these multiple-resolution methods, Rijthoven et al. developed the HookNet in 2020 for breast tissue segmentation [16]. HookNet consists of two U-Net style networks processing patches at different resolutions. Intermediate outputs from the network corresponding to the lower resolution were concatenated onto the output of an intermediate layer on the network corresponding to the higher resolution. This differs from the Multi-Resolution Network [14] because concatenation is chosen to be where the true image resolutions between the two networks align, rather than concatenating between symmetric parts of the encoder and decoder sides of the model. HookNet outperforms Multi-Resolution Net on the Radboudumc test set of breast cancer tissue types with a F1 score of 0.91 compared to 0.83. Due to HookNet's superior performance compared to Multi-Resolution Net [14] and U-Net variants, we build on the HookNet [16] architecture by creating a variant that uses 3 U-Net style networks to process three resolutions for melanoma segmentation.

3. Dataset

This study has been approved by the institutional review board. We have identified 57 UCSF patients with invasive melanoma for segmentation training and testing. In particular, these are primarily biopsies, stages T1 to T4, diagnosed between 2004 and 2014. These cases are single, individual melanomas and do not include melanomas co-existing with nevi or desmoplastic melanomas. Hematoxylin and Eosin (H&E) stained slides of the specimens are digitized at $40\times$ magnification using a Leica Aperio whole slide scanner, with a final resolution of approximately 3960 pixels per millimeter, or a pixel size of $(15.68 \times 15.68)(\mu m)^2$. Identifiers associated with the pathology slides are removed at scanning. The images are annotated using QuPath software [17] to indicate regions of epidermis, invasive melanoma, non-melanoma dermis (fibrosis, inflammation, normal dermis), and negative space (air, stratum corneum). For the invasive melanoma label, we include mass forming tumors, infiltrative strands, and individual cells. Fine blood vessels and stroma, lymphocytes, and necrosis within the border of the tumor are labeled as invasive melanoma. Adnexal epithelium is labeled as epidermis.

The annotation is done by two board-certified dermatopathologists. The initial annotations are drafted by trained students and residents. The pathologists then revise the annotations to the cellular and pixel level. All annotated slides are finalized by the consensus of both pathologists. Pathology images and corresponding label images are extracted as pairs.

4. Method

Invasive melanoma, or melanoma in the dermis, and in-situ melanoma, or melanoma in the epidermis, are visually indistinguishable, but invasive melanoma has greater clinical importance. The methods in this paper tackle semantic

segmentation of invasive melanoma.

To tackle this problem, we have developed a two-stage system, as shown in Figure 2. The first stage consists of employing two semantic segmentation models, one for melanoma and one for epidermis segmentation, respectively. The second stage intersects negative epidermis and positive melanoma predictions, to create final invasive melanoma predictions.

The most straightforward way of detecting invasive melanoma is to create a target class for such cells, and build a model to predict them. We have empirically verified that such an approach performs worse as discussed in results. Consequently, we developed a two-stage system whereby the epidermis segmentation map is used to refine the melanoma segmentation map by only selecting melanoma pixels outside of the epidermis. Another benefit of using this two-stage approach, rather than multi-class segmentation, is that obtaining additional annotated epidermis training data is significantly easier than obtaining additional annotations for invasive melanoma.

WSIs are large, high-resolution images and cannot be used for training by modern neural networks given GPU memory limitations. Thus we divide the entire WSI into smaller, square patches for model training and testing.

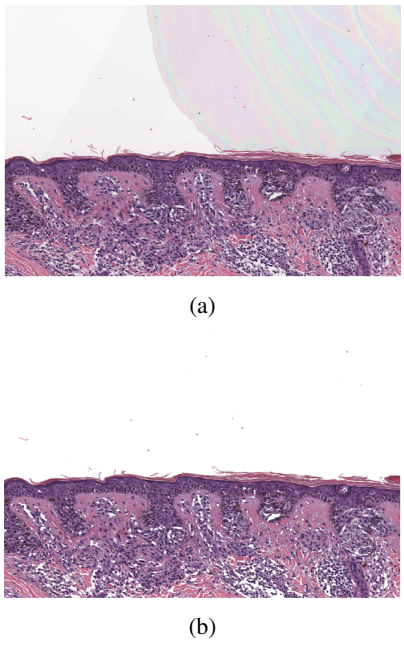
4.1. Data Augmentation

Our training dataset for the task of epidermis segmentation consists of 42 WSIs. Our data is heavily class imbalanced due to the relative sparsity of the epidermis in an entire WSI. Our training data exhibits an imbalanced pixel distribution of approximately 1:4 between the epidermis and negative class. Our experiments revealed that models trained on such a pixel distribution inevitably lead to underestimation of the epidermal regions while retaining a high accuracy because of the severity of the class imbalance.

To overcome this data imbalance in the training set, we alter the pixel distribution using sampling techniques [10]. Specifically, we remove patches that contain greater than 98% of pixels corresponding to regions without any tissue, because these patches are identical and uninformative for epidermis segmentation. We further undersample patches corresponding to dermal regions to increase the relative proportion of epidermis. Lastly, we oversample patches with large areas of epidermis by duplicating patches.

In addition to the aforementioned sampling modifications, we downscale the input patches to provide a larger window to the model. Specifically, we apply the detail-preserving image downscaling algorithm [18] to decrease the resolution of our input patches. This also combats class imbalance by decreasing the proportion of patches that lack a significant amount of target pixels.

Lastly, we apply a random combination of flips and rotations in multiples of 90 degrees to increase diversity in the training set. The resulting pixel distribution between the epidermis and negative class after these augmentations is approximately 3:7.



(a)

(b)

Figure 3: Glue removal of a WSI **(a)** With glue stains **(b)** Example of Otsu Thresholding to remove glue stains.

We use 57 WSIs in the training set for the melanoma segmentation task. We experiment with two distinct training approaches for our melanoma segmentation model shown in the lower branch of Figure 2. Specifically, we train the model with (a) invasive melanoma as the target class, and (b) with the union of invasive melanoma and epidermis as the target class. Since we combine the output of either model with the epidermis segmentation in the upper branch of Figure 2, logically, they should both result in the desired entity, invasive melanoma. We apply the same aforementioned sampling and data augmentation techniques to both approaches resulting in a pixel distribution 4:6 between melanoma and the negative class.

After the H&E staining process WSIs may contain artifacts such as adhesive stains. In an image containing any such artifacts we apply Otsu's thresholding technique to remove these artifacts from the input image before training, as shown in Figures 3a and 3b.

4.2. Network Architecture

Many deep learning approaches in biomedical imaging apply variants of the U-Net based architecture. U-Net based architecture is used by [9] and [10] for melanoma segmentation, and by others for similar tasks such as breast tissue segmentation [8, 9, 10, 14, 19, 20]. In this paper, we opt to use HRNet+OCR [20], a state-of-the-art segmentation

model developed in 2020. Specifically, [20] developed Object Contextual Representation(OCR) by substituting HRNetV2, a high-resolution network, into the modular backbone of the OCR architecture to arrive at HRNet+OCR. To leverage transfer learning we utilize weights from an instantiation of HRNet+OCR pretrained on the CityScapes dataset [21]. The pretrained HRNet+OCR was modified for binary segmentation, by adapting the segmentation heads of the output and auxiliary layers, with new parameter weights drawn from a uniform distribution.

In addition to HRNet+OCR, we also experiment with HookNet [16], a multi-resolution network that utilizes multiple image resolutions for segmentation of WSIs. This model takes multiple inputs of the same size, but at different resolutions centered on a given pixel. This provides more information for segmentation, albeit at the cost of lower resolution. We build on [16] by extending the HookNet model from two input resolutions to three. In our experiments, the two branched HookNet uses $40\times$ and $10\times$ magnifications, while the three branched HookNet uses $40\times$, $10\times$, and $5\times$ magnifications.

Furthermore, we perform experiments with the standard UNet architecture to serve as a baseline, and apply the FCN variant designed by [10] on our proposed dataset to adequately determine the effectiveness of our method.

4.3. Training Details

While training the HRNet+OCR models our goal is to minimize a weighted cross-entropy loss between the ground truth and predictions. The weight for a given class is the reciprocal of the square root of the class frequency or the reciprocal of the class frequency. To improve the feature representation of the input in the network, a term is added to the loss function corresponding to the cross-entropy loss between the ground truth and predictions from an auxiliary layer with weight 0.4 [20]. The auxiliary layer is an intermediate layer that also outputs predictions. Our overall loss function is:

$$\mathcal{L}_{overall} = 0.4 * \mathcal{L}(y_{aux}, y_{true}) + 0.6 * \mathcal{L}(y_{final}, y_{true})$$

where $\mathcal{L}$ is the standard cross-entropy loss function, y_{true} are the true labels, and y_{aux} and y_{final} are the softmax predictions from the auxiliary and final layers of the network, respectively.

We use the Adam optimizer and set the learning rate to 0.001 and beta parameters of Adam to 0.99 and 0.999. Both the melanoma and epidermis HRNet+OCR segmentation models are trained for 100 epochs.

The HookNet models use a cross-entropy loss between targets and predictions of each branch. Each of these losses are weighted using fixed parameters to construct the overall loss, with weights of 0.6, 0.4, and 0.3 for the $40\times$, $10\times$, and $5\times$ branches, respectively. We set the $l2$ regularizer λ parameter to 0.0001 and use the Adam optimizer. Both the melanoma and epidermis HookNet segmentation models are trained for 150 epochs.

We also utilize UNet and the FCN variant proposed in [10] as baseline methods for comparison. Our UNet models use the standard cross-entropy loss for each pixel, where each class was weighted using the same schema as utilized for HRNet+OCR. All the UNet models and the model proposed by [10] were trained for 150 epochs using an Adam optimizer with the same hyperparameters previously stated. We note that while training the model proposed by [10] our loss stagnated, so we decreased the learning rate by a factor of 10 to facilitate learning.

To train our models we use two GPUs: an NVIDIA GeForce GTX 1080TI and an NVIDIA Quadro M6000. This allows for larger batch sizes and larger patch sizes to be used for our model. Given GPU memory limitations, the largest allowable patch size is approximately 788×788 using a batch size of 4.

4.4. Evaluation

To evaluate our model's performance on the test set we use the following common metrics for segmentation models: mean intersection over union (mIoU), frequency weighted intersection over union (fwIoU), F1 scores, sensitivity, and specificity.

5. Experiment Results

In this section we will discuss the experiment results for our proposed method.

5.1. Epidermis Segmentation

Table 1 shows the performance of the stage 1 epidermis semantic segmentation models, shown in the upper branch of Figure 2, on the epidermis test set, which consists of 7 whole slide images. The highest mIoU (F1 score) achieved is 0.79 (0.74), as shown in Table 1 row 2, resulting from the HRNet+OCR model with downscaling of the 1576×1576 patches to 788×788 patches. Our epidermis segmentation method is comparable to the epidermis segmentation results presented in [10]. Specifically, as shown in row 2 of Table 1, our method yields a sensitivity and specificity, of 0.89 and 0.99 compared to 0.90 and 0.98 in [10], respectively. In some cases, our results are visually identical to the ground truth, as shown in Figure 4. Given the large variability in epidermis presentations, there are some cases where our model performs comparatively poorly such as epidermis surrounding hair follicles. This behavior is expected as the model observes relatively few examples of epidermis presented in this way, and can be addressed with a training set that includes more adnexal structures. Another negative contributor to the epidermis segmentation performance

#	Model	Input Patch Size	mIoU	fwIoU	F1 Score	Sensitivity	Specificity
1	3-Branched HookNet	800 × 800	0.76	**0.97**	0.69	0.91	0.98
2	HRNet+OCR	1576 × 1576	**0.79**	**0.97**	**0.74**	0.89	**0.99**
3	HRNet+OCR	3152 × 3152	0.77	**0.97**	0.72	0.95	0.98
4	UNet	1600 × 1600	0.75	0.96	0.69	**0.98**	0.98

Table 1: Stage 1, Epidermis Segmentation Results

#	Model	Targets	mIoU	fwIoU	F1 Score	Sensitivity
1	HRNet+OCR	Invasive Melanoma	0.5	0.94	0.1	0.49
2	2-Branched HookNet	Invasive Melanoma	0.51	**0.97**	0.08	0.06
3	3-Branched HookNet	Invasive Melanoma	**0.57**	**0.97**	**0.27**	0.36
4	UNet	Invasive Melanoma	0.43	0.80	0.08	**0.78**
5	[10]	Invasive Melanoma	0.53	**0.97**	0.14	0.13

Table 2: Stage 1, Melanoma Segmentation Results

#	Stage 1 Epidermis Model	Stage 1 Melanoma Model	mIoU	fwIoU	F1 Score	Sensitivity
1	HRNet+OCR 1576	HRNet+OCR (IM)	0.58	0.97	0.29	0.28
2	HRNet+OCR 1576	3-Branched HookNet (IM)	0.60	0.97	0.35	0.36
3	HRNet+OCR 1576	UNet (IM)	0.47	0.86	0.13	**0.74**
4	HRNet+OCR 1576	[10] (IM)	0.55	**0.98**	0.20	0.13
5	HRNet+OCR 1576	HRNet+OCR (IM ∪ E)	0.61	0.97	0.39	0.42
6	HRNet+OCR 1576	3-Branched HookNet (IM ∪ E)	0.58	0.97	0.32	0.50
7	HRNet+OCR 1576	UNet (IM ∪ E)	0.58	0.96	0.31	0.56
8	HRNet+OCR 3152	HRNet+OCR (IM)	0.56	0.97	0.24	0.22
9	HRNet+OCR 3152	3-Branched HookNet (IM)	0.62	**0.98**	0.41	0.37
10	HRNet+OCR 3152	UNet (IM)	0.49	0.89	0.15	0.18
11	HRNet+OCR 3152	[10] (IM)	0.56	**0.98**	0.22	0.14
12	HRNet+OCR 3152	HRNet+OCR (IM ∪ E)	**0.64**	**0.98**	**0.44**	0.39
13	HRNet+OCR 3152	3-Branched Hooknet (IM ∪ E)	0.62	0.97	0.41	0.37
14	HRNet+OCR 3152	UNet (IM ∪ E)	0.60	0.97	0.37	0.53
15	UNet	UNet (IM)	0.47	0.86	0.13	**0.74**
16	UNet	UNet (IM ∪ E)	0.57	0.97	0.27	0.29

Table 3: Stage 2, Invasive Melanoma Segmentation Results; the number after the epidermis model architecture indicates the patch size which was downscaled before being passed as input to the model; IM and E stand for invasive melanoma and epidermis, respectively, and denote the training targets of the melanoma model.

is the stratum corneum, which in some areas have an indistinct border with the epidermis. However, segmentation inaccuracy between the epidermis and the stratum corneum does not have any practical impact. The performance of our stage 1 epidermis segmentation model, shown in Table 1, indicates it could be used as a competitive model outside of the melanoma segmentation problem in applications such as [22, 23]. Table 1 also shows that both HookNet and HRNet+OCR are superior in almost every metric, and we further see when comparing row 4 to row 2 of 1 that transitioning from UNet to HRNet+OCR leads to a 5% and 7% gain in mIoU and F1 score, respectively.

5.2. Melanoma Segmentation

Table 2 shows the results of stage 1 melanoma semantic segmentation models, pictured in the lower branch of Figure 2. Stage 2 produces the invasive melanoma predictions, corresponding to the final output of Figure 2, and corresponding results can be found in Table 3. We test our two-stage approach on an invasive melanoma test set of 13 WSIs. The highest performing result, shown in row 12 of Table 3, uses (a) the HRNet+OCR model which downscales patches from 3152×3152 for epidermis segmentation, in conjunction with (b) another HRNet+OCR model trained using the union of invasive melanoma and epidermis as the

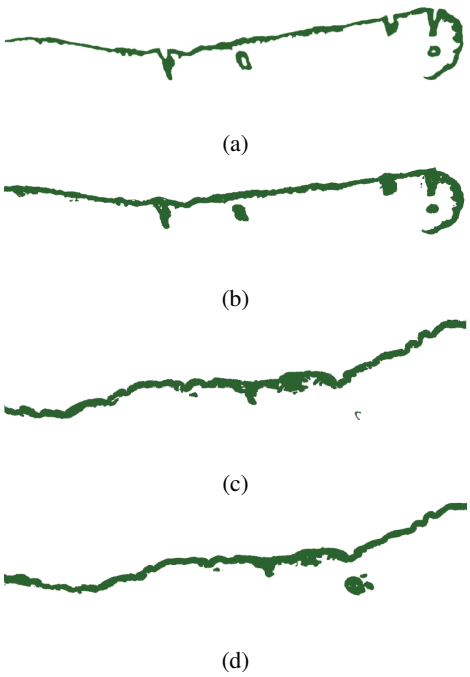


Figure 4: **(a)** and **(c)** are ground truth annotations corresponding to **(b)** and **(d)**, epidermis model outputs generated by the stage 1 HRNet+OCR model which downscales patches from 3152×3152.

targets. This approach yields a mIoU of 0.64, fwIoU of 0.98, and a F1 score of 0.44. We will compare this against the state-of-the-art later in this section.

Tables 2 and 3 demonstrate that combining melanoma predictions with epidermis segmentation models consistently increases the performance in almost every metric. This increase is nontrivial with gains of up to 16% in mIoU. This can be seen comparing stage 1 invasive melanoma results from Table 2, row 1 with mIoU of 0.5 to the stage 2 results in Table 3, row 1 with mIoU of 0.58, with both using the same melanoma model.

The best two-stage melanoma predictions result from using a stage 1 melanoma segmentation model with the union of invasive melanoma and epidermis as targets, as shown in Table 2. For instance, in rows 1 and 5 of Table 3, when the epidermis model is kept constant, changing the melanoma model target from invasive melanoma to the union of invasive melanoma and epidermis, mIoU increases from 0.58 to 0.61 and F1 score increases from 0.29 to 0.39.

In addition, we observe that using HRNet+OCR with patches downscaled from 3152×3152, in our two-stage method yields higher mIoUs, fwIoUs, and F1 scores compared to epidermis models using 1576×1576 patches in $3/4$ cases, indicating the benefit of information from a larger neighborhood for the epidermis model. This is shown in rows 2-7 and rows 9-14 of Table 3,

Our system can successfully identify very small clusters or single cells of melanoma in the dermis, as seen in Figure 5a. However, a current limitation is detecting individual cells or small clusters of melanoma within areas of high background inflammatory cells or fibrosis. This is challenging as an inflamed dermis with immune cell infiltrates can obscure the histopathologic features of melanoma aggregates.

As expected, our model performs better with large melanoma tumors, and predicts melanoma in the general area even if the predicted melanoma cell boundaries are not perfect. We also some observe false positives and false negatives at the boundary of the epidermis and dermis due to the process by which we join the two segmentation maps. The false positives can be removed by postprocessing to ignore melanoma regions directly connected to the epidermis in the segmentation output.

To the best of our knowledge, there are no published results comparable to semantic segmentation on our dataset, which includes fine clusters of melanoma. As seen in row 5 of Table 2 and row 12 of Table 3, our best model significantly outperforms [10] on our dataset with an approximately 21% gain in mIoU and 214% gain in F1 score. In addition, our best model also significantly outperforms a baseline UNet, shown in row 4 of Table 2, with performance improvements in lower mIoU, fwIoU, and F1 Score; these improvements are nontrivial with gains 49%, 22.5%, and 450% respectively.

5.3. Network Comparison

Throughout our experiments we use both HookNet, HRNet+OCR, UNet, and the model proposed by [10]. The 3-branched HookNet and HRNet+OCR obtained similar results for melanoma segmentation using the two-stage system with HRNet+OCR marginally outperforming HookNet performance. Specifically, as seen in rows 12 and 13 in Table 3, HRNet+OCR outperformed the 3-branched HookNet's mIoU, fwIoU, and F1 score by 3%, 1%, and 7%, respectively.

We also observe that UNet underperforms in most metrics relative to models such as HookNet and HRNet+OCR, which partially corroborates the observations of [16]. UNet does attain the highest sensitivity as observed in Figure 5b and the rest of our test set; this is because the model produces many false positives which consequently yields lower precision. In particular, we observe that in rows 1-5 of Table 2, where all models are trained with the same targets, UNet had the lowest mIoU and F1 score of 0.43 and 0.08 compared to the maximum values of 0.57 and 0.27, attained by the three branched HookNet. UNet also attained the minimum mIoU of 0.47 when utilized in our two stage method, as shown in rows 3 and 15 of Table 3. Furthermore comparing row 4 of Table 2 to row 10 of Table 3 shows that

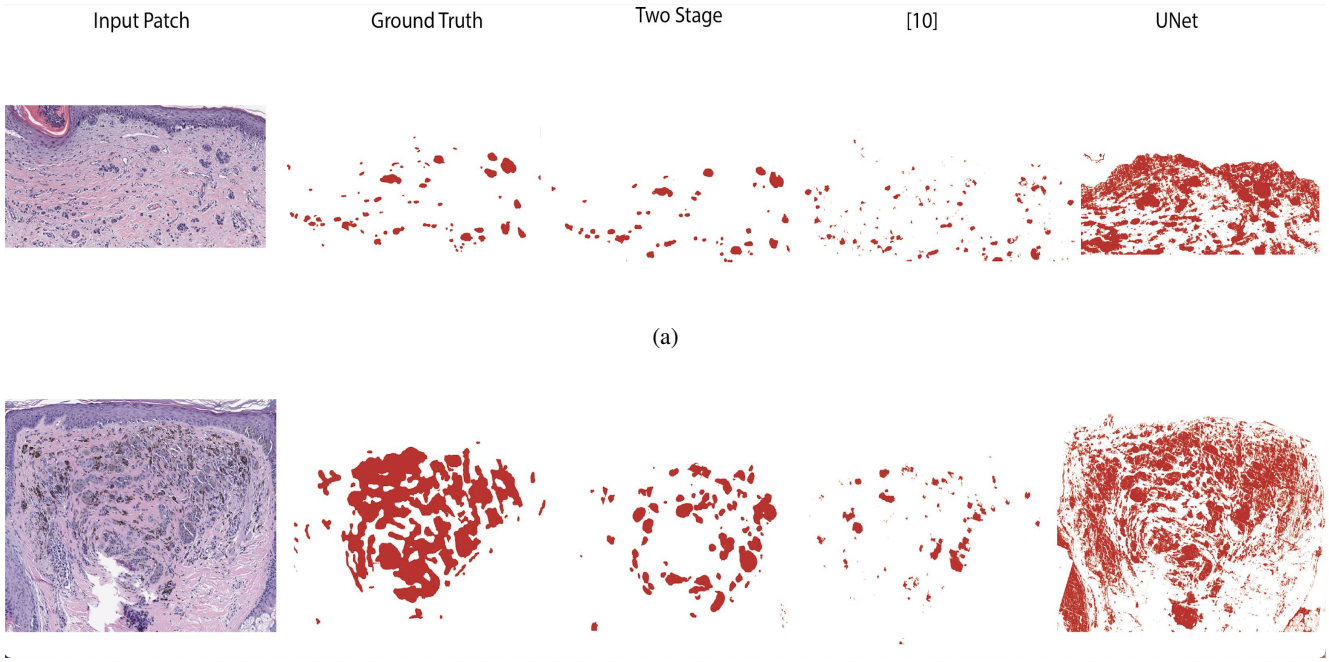


Figure 5: Two examples of predictions from the two-stage model from row 12 of Table 3, FCN model variant in [10] from row 5 in Table 2, and 2 stage UNet from row 3 in Table 3 respectively; **(a)** model predictions on a WSI patch containing a hair follicle; **(b)** model predictions on a WSI patch without an anomalous skin structure. For the second row, the full WSI mIoUs of the two-stage approach, FCN model variant in [10], and 2 stage UNet are 0.630, 0.523, and 0.518 respectively

utilizing UNet in our two stage method leads to a nontrivial gain of more than 10% in mIoU. These two results indicate that recently proposed networks such as HookNet and HRNet+OCR are more suitable in biomedical imaging than UNet. More importantly, regardless of the model, the two stage model in row 12 Table 3 significantly outperforms single stage traditional models for our particular application.

Furthermore, the FCN model variant in [10] for melanoma segmentation in the two stage approach, as shown in rows 4 and 11 of Table 3, does not offer improvements over the 3-Branched HookNet or HRNet+OCR models. Utilizing HRNet+OCR 1576 as the epidermis segmentation model in the two stage approach with [10] attains a mIoU of 0.55 and F1 score of 0.20; utilizing HRNet+OCR 3152 attains a mIoU of 0.56 and F1 score of 0.22. Our best model in row 12 of Table 3 significantly outperforms these results by 14% in mIoU and 100% in F1 score.

6. Conclusions and Future Work

We propose a novel, two-stage pipeline to markedly improve segmentation of invasive melanoma at the pixel level in whole slide pathology images. The first stage segments the tissue into (a) epidermis regions and non-epidermis regions, (b) the melanoma and non-melanoma regions. The second stage combines the two results from the first stage to yield the invasive melanoma predictions, which by definition is outside of the epidermis, and is of significant clinical importance. The performance of our models on the test set show that our method is able to produce superior segmentation results on most melanomas encountered clinically, including smaller tumors not present in previously published works.

The next research steps include testing this hypothesis with 80% power at 5% significance level. This will require 197 test samples assuming the standard deviation of the IoU is 0.5. The deep learning method and manually measured Breslow thickness will be compared. The calculated cross-sectional area of invasive melanoma will be used to develop a survival prediction algorithm using the deep learning calculated area.

In the future, we will extend this study to create a tool to assist medical professionals with end-to-end processing and analysis of medical data. This tool could lead to improvement in prognostic prediction and management guidance for melanoma patients, as well as revision of the T-category of the staging system. There is also potential for this method to be generalized to other invasive malignancies, especially carcinomas that have in-situ and invasive components.

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